

Matthew Doyle

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PROJECTS

Institutional Order-Flow Equity Strategy

2026–Present

Python, pandas, scikit-learn, XGBoost, SQL, DuckDB, DuckLake, IBKR

- Developed a medium-horizon, dollar-neutral long-short strategy for S&P 500 equities using proprietary order flow data from ExTech.
- Implemented data pipeline for ingestion, warehousing, feature engineering, model training, portfolio construction, backtest and paper execution.
- Produced three-year out-of-sample backtest with a 13.9% ANR, 1.61 Sharpe ratio and 6.1% maximum drawdown after estimate transaction costs.
- Developed statistical risk model using PCA and hedge portfolio to manage systematic factor exposure, producing approximately zero market beta.

Prediction Market Binary Option Arbitrage

2026–Present

Python, Redis, data engineering, Databento, trading APIs, trade execution

- Built research and automated execution system to evaluate binary-option arbitrage opportunities across prediction market and traditional venues.
- Collected and normalized market and order-book data to identify zero theoretical risk opportunities with a multi leg options strategy.
- Automated simultaneous order execution, fill monitoring and position reconciliation and execution risk mitigation through trading APIs.
- Deployed the strategy with live capital, generating an annualized money-weighted return of approximately 40% since inception.

PROFESSIONAL EXPERIENCE

The Entrepreneurship Hatchery, University of Toronto

Toronto, Ontario

Venture Associate (contract)

Sep 2025–Mar 2026

- Sourced dealflow for the accelerator by screening technology projects and founders in the University of Toronto engineering research community.
- Conducted commercial diligence on early-stage technology ventures, assessing market opportunity, positioning, and team/founder fit.
- Planned and delivered community programs connecting researchers, founders, mentors and investors across the University of Toronto ecosystem.

Higher Education Strategy Associates

Toronto, Ontario

Research Analyst

May 2024–Sep 2025

- Consulted on academic-program viability through analysis of enrolment, labour-market demand, industry hiring patterns and program curricula.
- Analyzed provincial funding programs and institutional financial data to support client recommendations and strategic planning.
- Conducted expert interviews and skill-gap research to support curriculum strategy and institutional partnerships.

Students Nova Scotia

Halifax, Nova Scotia

Chair, Board of Directors

May 2023–Apr 2024

- Oversaw board governance and organizational strategy, liaising with member organizations, government officials and external stakeholders.
- Helped secure over \$16 million in financial-aid grants through policy research, government advocacy and stakeholder engagement.

St. Francis Xavier Students' Union

Antigonish, Nova Scotia

Vice President, External Affairs

May 2023–Apr 2024

- Led research and advocacy on student housing, financial aid and education policy, developing recommendations for government stakeholders.
- Advanced reforms enabling post-secondary institutions to access CMHC financing for student housing and institutional development.
- Represented students before provincial and national policymakers while coordinating government relations and sector partnerships.

EDUCATION & CREDENTIALS

St. Francis Xavier University

Antigonish, Nova Scotia

Bachelor of Arts, First Class Honours with Distinction, Philosophy

Sep 2019–May 2024

- Wrote an honors thesis on the normative and applied ethics of the alignment problem in artificial intelligence.
- **Awards and Scholarships:** Dean's List (2023, 2024), Father Charles R. MacDonald Memorial Medal in Philosophy (2024), Mulroney Institute of Public Policy and Governance Case Competition (1st place 2024), Students Nova Scotia Most Valuable Player Award (2024), Canadian Alliance of Student Associations Advocacy Expert Award (2024), Golden X Leadership Award (2024)

Credentials

CFA Program — Level I Candidate, Nov 2026

Python for Data Science — University of Michigan

Options 101 — Akuna Capital

SKILLS

Technical: Python, pandas, NumPy, scikit-learn, XGBoost, SQL, DuckDB, DuckLake, Docker, Google Cloud CLI/SDK, OpenAI API, Notion, Vercel and LaTeX.

Quantitative research: Backtesting, order-flow analysis, market-data pipelines, signal research, risk modelling, portfolio construction, paper-trading execution and model evaluation.

General research: Financial analysis, investment memos, market and policy research, literature reviews, market sizing and expert interviews.